Kirirom Institute of Technology

Operations Research

– Historical development –

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Operations Research

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Branch of applied mathematics used in scientific decision making --> **OPTIMUM DECISION**

Developed during the second world war to solve strategic and tactical issues using scarce resources to achieve objectives.

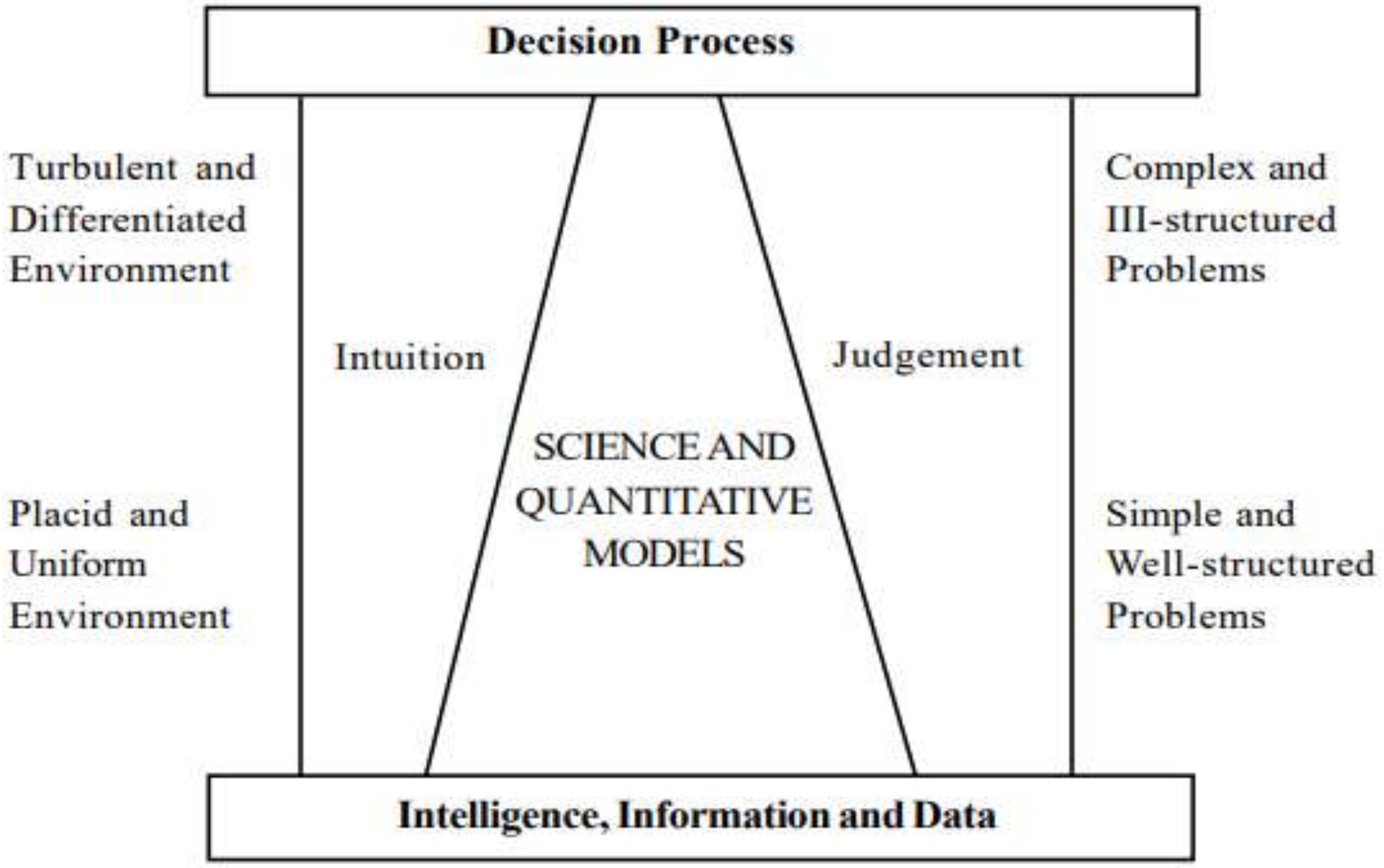
OPERATIONS ➡ military operations
RESEARCH ➡ new methods of doing things

Scarcity of raw materials
Low production output
Recession

➡ Optimal solution

⬇

Development of Linear Programming



Operations Research was successful developed in the British Army
The United States Army started implementing Operations Research
Subsequently it spread to the industry

Strategic = Plan for achievement of long term goals/vision
Tactic = Short term actions to achieve the strategy

DECISION = PROCESS OF CONCLUSION

Decision involves

- Goals or objectives
- Set of alternatives
- Criteria for choosing a course of action

Phases of decisions:

1. Formulation of goals and objectives, enumeration of environmental constraints, identification and evaluation of alternatives.

2. Selection of optimal course of action for a given set of constraints

OR is concerned with choosing an optimal strategy under specified set of assumptions

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Classification of decisions

I Decisions (depending on the purpose)

- **Strategic**
  Related to external Environment
- **Administrative**
  Resource utilisation of an orgnisation.
- **Operational**
  Related to day –to-day Repetitive work.

II Decision (Depending on the nature)

- **Programmed decisions**
  Meant for repetitive and well-structured problems. Inventory Problems, Product Mix Problems, etc.
- **Non Programmed decisions**
  Meant for non-routine, novel, ill-structured problems. Policy matters, Product market mix, plant location Etc.

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Classification of decisions

III Decisions (Depending on the persons involved)

- Individual
- Managerial

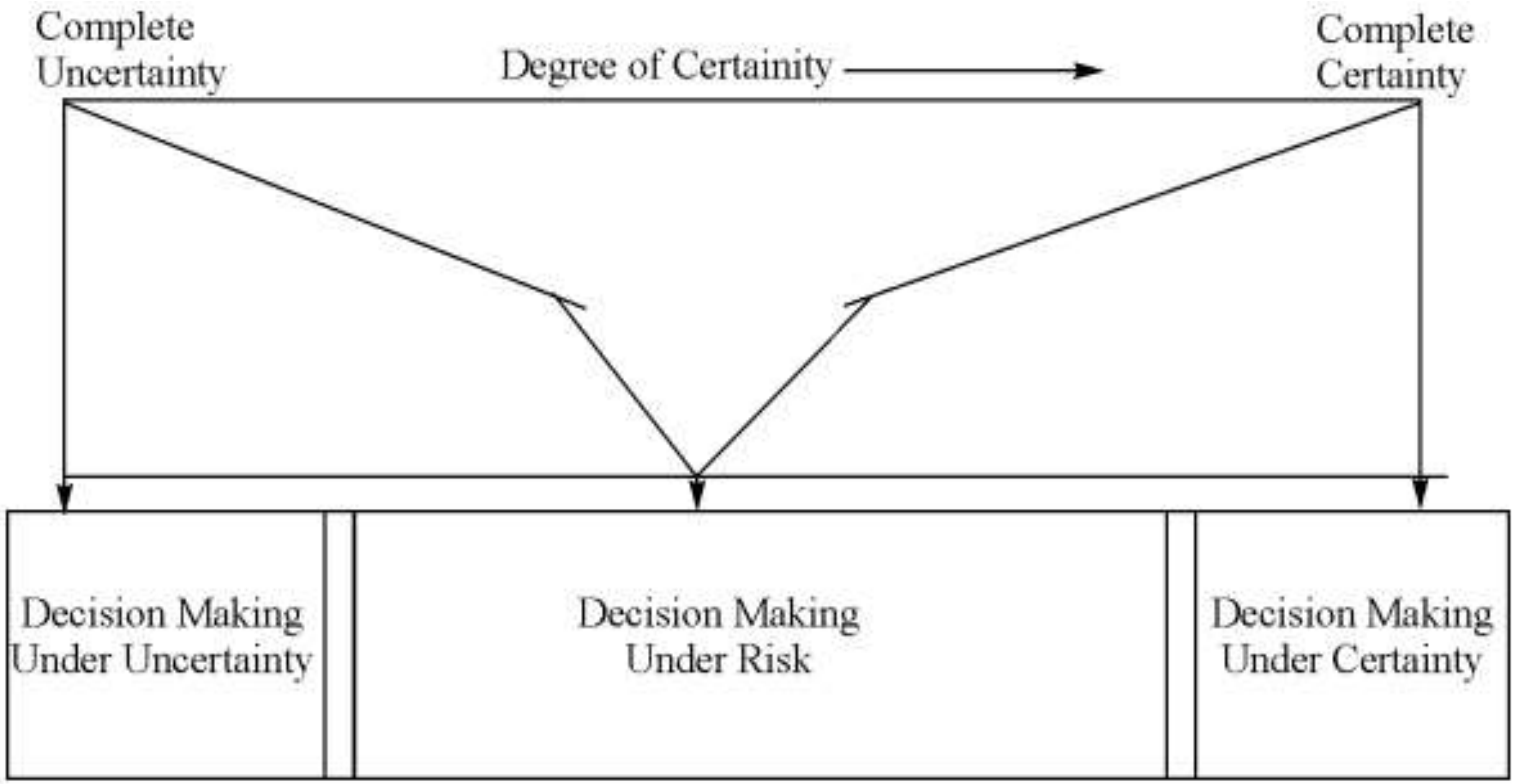
IV. Decisions (Depending on the Sphere of interest)

- Economic
- Social
- Business
- Political
- Social

V. Decisions (depending on the time horizon)

- Static (One decision for entire planning period)
- Dynamic (Decisions are sequential)

Decision based on certainty



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Optimal Decision

To provide a scientific basis to the decision maker for solving problems involving interaction of various components of an organization by employing a team from various disciplines, all working together in finding a solution which is in the best interest of the organization as a whole.

The best solution thus obtained is “optimal decision”

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Characteristics & scope of Optimal Decision

Characteristics:

Involves an interdisciplinary team

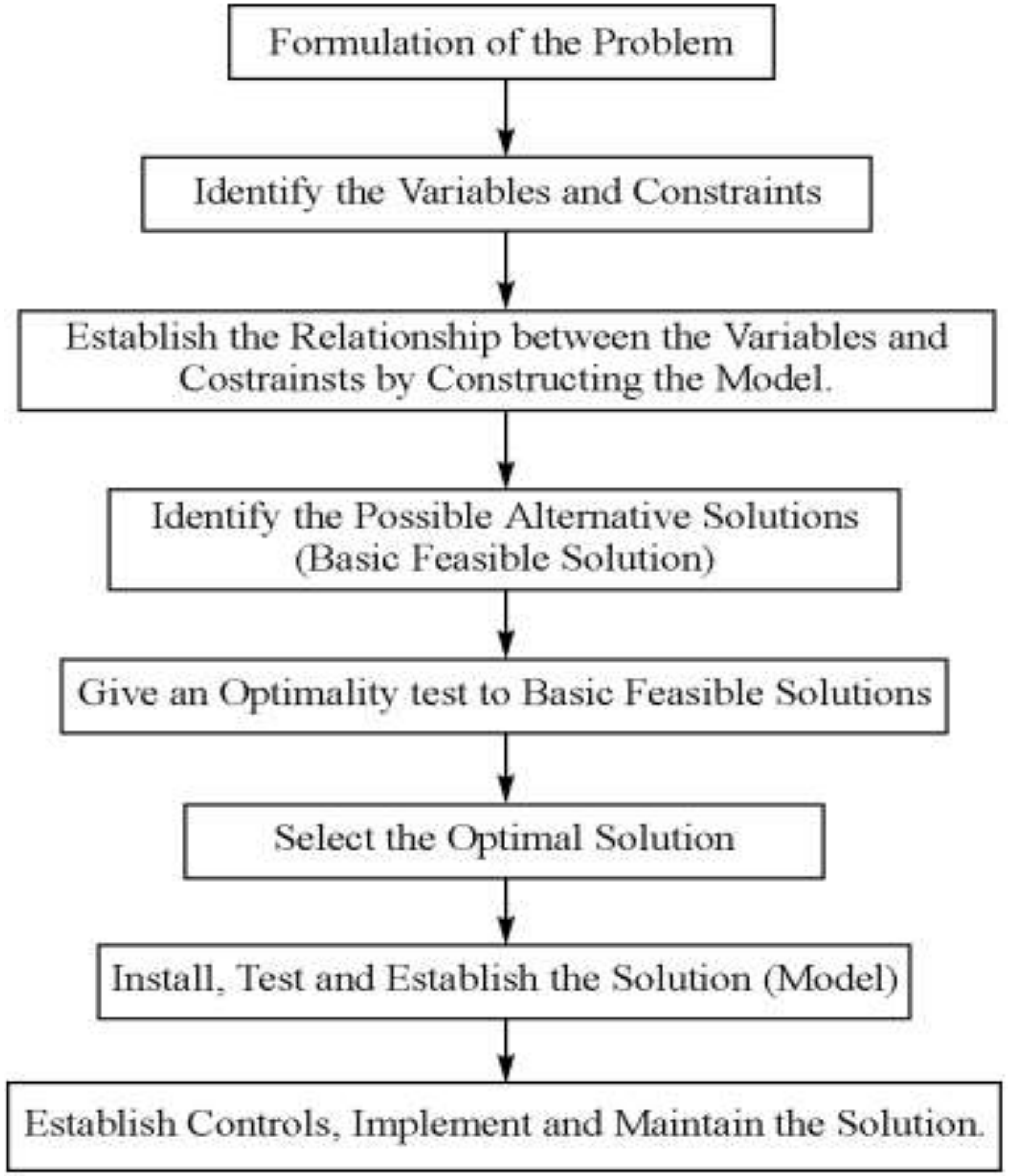
Enhances ability of decision maker in using various mathematical tools & models

A system based approach

Scope:

Decision making in the areas of
Defense operations (Game theory, linear programming, inventory models)
Industry (decision models, inventory models, linear programming, transportation model, sequencing, assignment model, and replacement models)
Economic development (planning)
Agriculture (optimizing crop yield)
Traffic flow & control (queuing theory)
Healthcare services (managing beds, medicine shortages, etc.)

Steps in solving OR problems



Terms in Operations Research

Variables	:	Variables are used to find out minimum quantities of two products X and Y (in capital letters)
Constraints	:	Limitations and restrictions
Model	:	Representation of a situation (represented by x and y (in small letters)
Objectives	:	Something that one plans to achieve
Goals	:	An achievable outcome that is broad and long-term Goals are SMART (Specific, Measurable, Attainable, Realistic and Time bound).

Model building

1. IDENTIFYING VARIABLES & CONSTRAINTS

2. ESTABLISHING A RELATIONSHIP AMONG VARIABLES & CONSTRAINTS

Example of a model using one machine A:

If a company manufactures x units of item X and

y units of item Y,

1 unit of x consumes 1 hour in machine A and

1 unit of y consumes 1 hour in machine A,

then the total consumption of x and y in machine A is given by

$$\mathbf{1x + 1y = A}$$

Model building

Example of a model with 3 machines:

Machine A has maximum capacity 40 hours
Machine B has maximum capacity 240 hours
Machine C has maximum capacity 350 hours

1 unit of product X consumes
1 hour of machine A
3 hours of machine B
10 hours of machine C

1 unit of product Y consumes
1 hour of machine A
8 hours of machine B
7 hours of machine C

Profit contribution of X is 5
Profit contribution of Y is 7

The capacity of machine A will be 40, then $1x + 1y \leq 40$

The capacity of machines B will be $3x + 8y \leq 240$

The capacity of machine C will be $10x + 7y \leq 350$

Structural constraint

The profit contribution will be $5x + 7y$

If we have to maximize profit, then the objective function is

Maximize $Z = 5x + 7y$

Objective function

where x and y are ≥ 0

Nonnegative constraint

Broad classification of models

Mathematical & descriptive model:

Descriptive model:

Provides description about a system giving variables, constraints and objectives of the system or problem.
Has a limited use in operations research

Mathematical model:

Explains the entire system in mathematical language.
Helps the operations researcher in arriving at a solution

Static & dynamic model:

Static models do not change
Dynamic models are constantly changing

Classification of models

Models are classified based on structure, purpose, nature of environment, behavior, methods and use of digital computers

Classification by structure:

Iconic model:

Scaled version of actual objects.
Eg: Scaled toy of a car, model of an internal combustion engine
Easy to construct, can explain the static or dynamic behaviour of the system

Analogue models:

One set of models are used to represent another set of properties
Eg.: Blue color represents water, white represents cloud, contour lines on map, graphical representations

Symbolic models or mathematical models:

Variables are represented by mathematical symbols
Eg.: Resources allocation

Classification by utility

Descriptive model:

Describes the problem (certain aspects of problem) in a system

Predictive model:

Can predict the results based on data collection

Prescriptive model:

If the predictive model is successful, it can be used for finding a course of action/ solution to a given problem to achieve a goal

Classification by nature or environment

Deterministic model:

There is complete certainty about the values of the variables, available resources and expects that they do not change during the planning process.
Does not contain elements of uncertainty or probability
Eg: Linear programming model

Probabilistic or Stochastic model:

There is an element of probability and as a result, the values of variables and outcomes of certain actions cannot be predicted.
There is an element of risk involved
The degree of certainty varies from situation to situation.
Eg.: Sale of insurance policies

Stochastic = Pattern that may be analyzed statistically but cannot be predicted

Classification based on obtaining the solution

Analytical models:

This model has a well defined mathematical structure
Can be solved using mathematical techniques

Simulation models:

This is an imitation model
It has a mathematical model, but cannot be solved using mathematical techniques
It requires experimental analysis
Complex systems can be studied using simulation

Points to remember while model building

1. Solve a situation using a simple model, do not try to complicate.

2. Before building a model, understand and clarify its purpose.

3. Test and validate a model before implementing.

4. A model has to be used based on the purpose it is meant for.

5. Model is a guide to decision making. Model cannot replace a decision maker.

6. A model should be as simple as possible.

7. A model should be as accurate as possible.

8. The benefit of model depends on the process by which it is developed.

Thank You!